

2026 National Canadian Medical Datathon

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Dataset Overview- BUPA Liver Disorders

Sample: 341 adult male participants (n = 345 - nulls and duplicates)

Variables:

- Drinks: # half-pint equivalents of alcohol consumed/day
- Alanine Aminotransferase (ALT): 17-63 IU/L (N)
- Aspartate Aminotransferase (AST): 15-37 IU/L (N)
- Alkaline Phosphatase (ALP): 50-136 IU/L (N)
- Gamma-Glutamyl Transpeptidase (GGT): 15.0-85.0 IU/L (N)
- Mean Corpuscular Volume (MCV): 80-100 fL(N)

Identifying Alcoholic Liver Disease

As per criteria from the American College of Gastroenterologists (2024), common indicators of alcoholic liver disease include:

Patient History

Patient history of “chronic alcohol use” is one of the strongest identifiers of alcoholic liver disease

Clinical Exam Findings

Constitutional symptoms, nausea, jaundice, ascites/edema, hepatic encephalopathy, GI bleeding, fever, hepatosplenomegaly etc.

Lab indications

AST:ALT Ratio > 1.5
AST and ALT Values <400 IU/L
GGT elevated (sometimes),
elevation not specific to
alcoholic liver disease

THE PROBLEMS WITH DIAGNOSIS

Liver enzyme
abnormalities are often
clinically misinterpreted



Delayed intervention
Inappropriate referrals
Healthcare inefficiencies

Reactionary
medical treatment for
abnormal blood tests



Missed prevention
opportunities

Patients access
test results alone on
HealtheNL

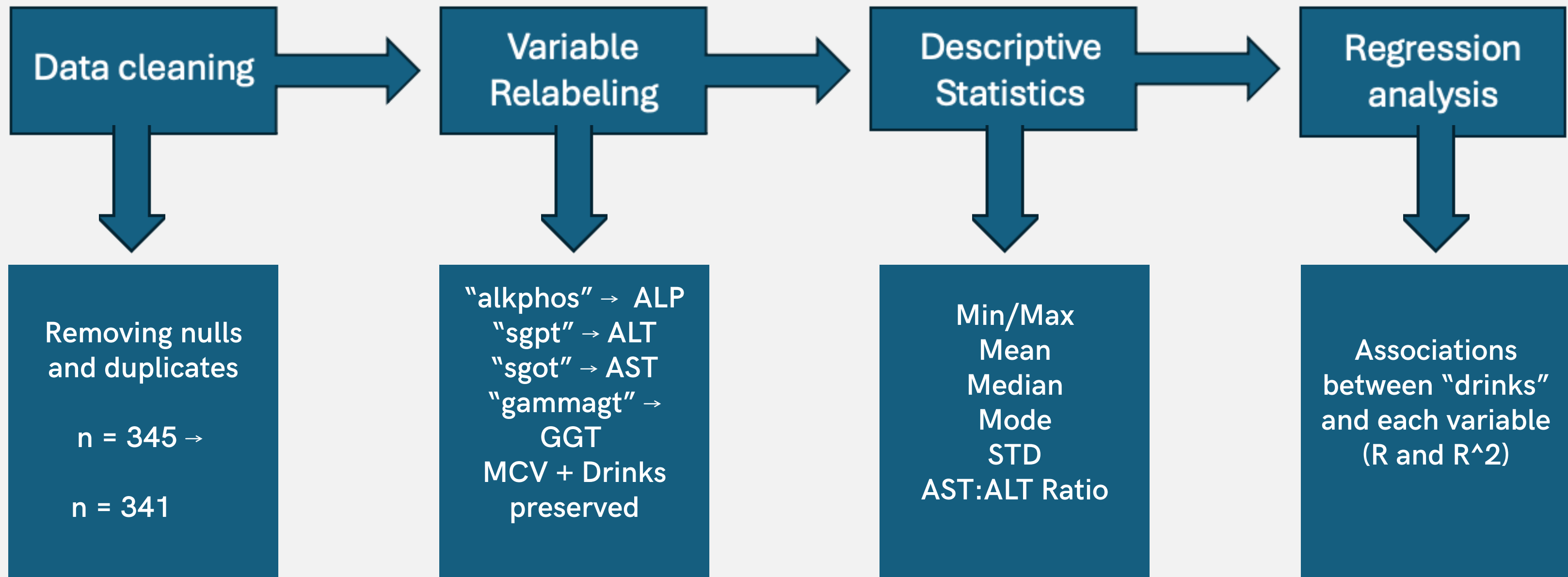


Patient confusion

RESEARCH QUESTIONS

1. Can we identify alcoholic liver disease using patient drinking patterns, liver enzymes and blood test patterns?
 2. Can we create an assessment tool for alcohol-related liver risk using these variables?
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Methods



VARIABLE	MEAN	MEDIAN	MODE	STD DEV	MIN	MAX	COUNT
ALP	69.89	67.00	62.00	18.43	23.00	138.00	341
ALT	30.51	26.00	17.00	19.59	4.00	155.00	341
AST	24.66	23.00	20.00	10.12	5.00	82.00	341
Drinks	3.43	3.00	0.50	3.34	0.00	20.00	341
GGT	38.40	25.00	14.00	39.44	5.00	297.00	341
MCV	90.12	90.00	91.00	4.45	65.00	103.00	341
Selector	1.58	2.00	2.00	0.49	1.00	2.00	341

Table 1. Descriptive statistics of variables ALP, ALT, AST, Drinks, GGT and MCV

Regression Analysis

Methodology:

To evaluate the relationship between alcohol consumption and liver injury, a linear regression analysis was performed using patient data containing drinks per day and key liver enzymes (ALT, AST, GGT, ALP).

For each enzyme, alcohol consumption was treated as the predictor, and the enzyme level as the outcome variable. The regression estimates how enzyme levels change as alcohol intake increases.

The analysis reports:

Slope – change in enzyme level for each additional drink per day

Intercept – estimated enzyme level at zero alcohol intake

Correlation (r) – strength of association between alcohol and the enzyme

R^2 – proportion of enzyme variability explained by alcohol consumption

Count Used – number of patient records analyzed

This analysis helps determine which liver enzymes are most associated with alcohol consumption, supporting the clinical reasoning used in the liver damage risk model.

Linear Regression Equation

$$Y = a + bx$$

$$a = \frac{[(\Sigma y)(\Sigma x^2) - (\Sigma y)(\Sigma xy)]}{[n(\Sigma x^2) - (\Sigma x)^2]}$$

$$b = \frac{[n(\Sigma xy) - (\Sigma x)(\Sigma y)]}{[n(\Sigma x^2) - (\Sigma x)^2]}$$

Enzyme = (Slope x Drinks) + Intercept

VARIABLE	SLOPE	INTERCEPT	CORRELATION (R)	R ²	COUNT USED
alp	0.551	68.000	0.100	0.010	341
alt	1.236	26.272	0.211	0.044	341
ast	0.850	21.747	0.281	0.079	341
ast_alt_ratio	-0.003	0.952	-0.025	0.001	341
ggt	4.058	24.479	0.344	0.118	341
mcv	0.412	88.707	0.309	0.096	341

Table 2. Regression analysis between variables ALP, ALT, AST, Drinks, GGT, MCV and AST:ALT ratio and Drinks

Dataset Limitations

- Age
- Distribution
- Sample size
- Chronicity
- Sex (males only)
- Comorbidities
- Diagnosis



CLINICAL RELEVANCE OF LIVER DISEASE DATA ANALYSIS

- Alcohol-related liver damage cannot be exclusively evaluated or screened for by abnormal liver enzymes → often normal in alcoholic liver disease
- Preventative healthcare based on risky lifestyle factors (e.g. alcohol consumption) is key
- Treat the person, not the numbers!



Liver Risk Assessment Tool

Purpose:

Create an educational tool that estimates alcohol-related liver risk (0–100 score) using:

- Age
- Drinks per day
- Liver enzymes: ALP, ALT, AST, GGT

Note: This tool is currently intended for educational purposes and does not have enough data backing to provide conclusive medical diagnoses

Niemela O, Niemela M, Bloigu R, Aalto M, Laatikainen T (2017) Where should the safe limits of alcohol consumption stand in light of liver enzyme abnormalities in alcohol consumers? PLoS ONE 12(12): e0188574. <https://doi.org/10.1371/journal.pone.0188574>

Liver Risk Assessment Tool

Source: Histogram gradient boosting

**Evaluation: 5 fold cross validation repeated 4 times (20 runs)
for reliable results**

**Performance: ROC-AUC 0.783, Accuracy 74%, F1 Score:
0.785**

**Key predictors: AST/ALT ratio (21.2%), GGT (19.6%),
Drinks/day (6.2%)**

DEMO

datathon-silk.vercel.app

CLINICAL RELEVANCE OF RISK ASSESSMENT TOOL



- Guideline application
- Faster time to diagnosis
- Increase Healthcare Efficiency
- Improves patient knowledge
- Refers to local resources
- Encourages lifestyle change

Conclusion

Summary

Liver enzymes alone are not effective in predicting alcoholic liver disease, and developed an educational clinical risk assessment tool to estimate potential alcohol-related liver damage.

Next Steps

The tool could be expanded to include disease parameters for other liver disorders to allow additional health education.

- Improve the model using larger clinical datasets with more variables, broader distribution with “sick” patients, and confirmed disease outcomes

Q & A
